

LONGERENONG, Australia

WMO: 958350

Lat: 36.6722S

Lon: 142.2992E

Elev: 133

StdP: 99.74

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.1	1.2	-5.0	2.5	23.4	-2.8	3.0	21.8	12.9	10.0	11.4	11.9	1.7	240	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.4	37.8	18.7	35.3	17.9	33.1	17.5	21.5	28.9	20.1	29.3	19.0	28.7	5.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.6	14.6	22.9	17.7	12.9	21.9	16.0	11.6	21.6	63.2	28.4	58.0	28.7	54.5	28.5	36.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	10.1	9.0	-2.2	42.6	1.0	1.7	-2.9	43.9	-3.5	44.9	-4.1	45.9	-4.9	47.1		
(5)			-2.3	25.1	1.0	5.2	-3.0	28.8	-3.6	31.8	-4.2	34.7	-4.9	38.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	21.9	21.7	18.7	15.1	11.6	9.2	8.7	9.1	11.0	13.4	17.1	19.4		
	DBStd	5.90	4.52	4.26	4.17	3.45	2.64	2.10	1.92	2.21	2.87	3.87	4.47	4.56		
	HDD10.0	174	0	0	0	2	12	39	49	43	19	8	1	0		
	HDD18.3	1753	14	14	47	109	208	275	299	285	222	162	77	40		
	CDD10.0	1886	368	326	269	153	62	14	8	17	49	113	213	292		
	CDD18.3	423	124	107	58	11	0	0	0	0	1	9	39	74		
	CDH23.3	6191	1803	1373	742	161	4	0	0	0	19	216	666	1207		
	CDH26.7	3058	983	691	324	42	0	0	0	0	2	78	306	633		
(14) Wind	WSAvg	4.4	5.0	4.9	4.7	4.0	3.9	3.7	4.1	4.1	4.2	4.4	4.6	4.9		
(15) Precipitation	PrecAvg	422	26	21	21	27	46	40	48	47	45	45	32	23		
	PrecMax	810	98	155	100	118	142	146	100	93	122	172	86	72		
	PrecMin	190	0	0	0	0	5	8	12	6	9	2	0	1		
	PrecStd	119	24	27	20	26	30	28	24	21	28	34	22	20		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.9	39.7	36.9	30.9	23.6	17.6	16.7	20.8	26.0	32.7	36.5	40.6		
		MCWB	19.5	20.0	18.1	17.1	13.9	12.1	11.2	12.9	14.0	15.2	18.8	18.2		
	2%	DB	38.2	36.1	33.3	27.4	20.7	15.6	14.8	17.6	22.3	29.0	33.3	36.6		
		MCWB	18.9	18.6	17.6	15.2	13.2	11.7	10.8	11.3	13.8	14.7	17.1	17.7		
	5%	DB	35.0	33.7	30.3	24.9	18.5	14.5	13.6	15.5	19.5	25.5	30.4	33.3		
		MCWB	17.8	18.4	16.9	14.6	12.6	11.1	10.4	10.8	12.6	13.8	16.7	16.9		
	10%	DB	32.2	31.0	27.5	22.5	16.8	13.4	12.6	14.0	17.2	22.3	27.2	30.1		
		MCWB	17.4	17.6	16.1	13.7	12.2	10.6	9.8	10.2	11.9	13.0	15.5	16.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.8	25.5	20.5	18.1	16.3	14.0	13.0	13.8	16.4	18.1	22.5	22.7		
		MCDB	29.9	26.8	28.7	25.9	18.3	15.2	14.2	17.7	21.6	26.5	25.4	30.7		
	2%	WB	21.0	21.7	19.2	16.7	14.6	12.6	11.7	12.4	14.7	16.1	19.1	19.5		
		MCDB	29.6	29.2	28.0	22.5	18.1	14.5	13.6	15.9	19.5	23.8	27.1	28.9		
	5%	WB	19.7	20.1	18.1	15.6	13.6	11.7	10.9	11.4	13.5	14.9	17.7	18.2		
		MCDB	31.2	27.7	26.9	22.1	16.9	13.6	12.7	14.6	18.1	22.5	26.5	28.5		
	10%	WB	18.5	18.9	17.1	14.6	12.7	11.0	10.2	10.5	12.3	13.8	16.4	17.0		
		MCDB	30.1	27.7	26.1	21.1	15.9	12.8	11.9	13.5	16.6	21.1	24.6	28.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.4	15.3	14.2	13.1	10.4	8.9	8.4	10.1	12.1	14.7	15.6	16.3		
		MCDBR	20.1	18.6	17.5	16.4	13.0	10.1	9.7	12.6	15.8	20.4	20.3	20.9		
		MCWBR	6.6	6.0	6.1	7.2	6.9	6.6	6.4	7.4	8.2	8.4	7.2	6.8		
	5% WB	MCDBR	15.9	14.1	14.8	13.4	10.1	8.5	8.4	11.0	13.4	16.1	16.6	16.2		
		MCWBR	6.9	6.3	6.2	6.7	6.2	6.3	6.1	7.0	7.9	7.8	7.9	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.338	0.333	0.316	0.309	0.301	0.292	0.292	0.304	0.322	0.331	0.337	0.330		
	taud		2.496	2.520	2.566	2.565	2.574	2.594	2.591	2.534	2.472	2.457	2.472	2.511		
	Ebn at Noon		996	978	958	905	854	835	856	896	935	969	990	1009		
	Edh at Noon		114	106	94	83	71	65	69	84	101	112	116	113		
(44) All-Sky Solar Radiation	RadAvg		7.70	6.85	5.43	3.85	2.59	2.06	2.22	3.14	4.34	5.79	6.82	7.56		
	RadStd		0.39	0.33	0.25	0.20	0.12	0.16	0.13	0.20	0.31	0.41	0.43	0.36		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		+0.43	N/A	N/A	+0.86	N/A	N/A	N/A	-87	+139	+59

Nomenclature: See separate page